

John S. Ho

I work on integrating physical technology with humans, across wearables and implants. I'm currently a research scientist on the Human AI team, where I lead the technical direction for sensing across our wearables — the sensors that let a device feel natural to use and understand the person using it.

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INDUSTRY EMPLOYMENT

- 2022-Present Research Scientist
Human AI, Meta Platforms, California
Developing sensors and algorithms for novel input and interaction.
- 2021 External Consultant
Medical Devices Group, Cambridge Consultants, Cambridge, UK
Prototyped solutions for connected biosensors and medical devices.

ACADEMIC EMPLOYMENT

- 2022-Present Associate Professor
Department of Electrical and Computer Engineering,
Institute for Health Innovation and Technology (affiliate faculty)
The N.1 Institute for Health (affiliate faculty)
National University of Singapore, Singapore
- 2015-2022 Assistant Professor
National University of Singapore, Singapore

EDUCATION

- 2015 PhD, Electrical Engineering, Stanford University, USA
Thesis titled "Wireless Powering for Bioelectronics" (Advisor: Ada S Y Poon)
- 2012 MS, Electrical Engineering, Stanford University, USA

AWARDS

- 2020 Young Scientist Award, Singapore National Academy of Science
- 2020 IES Engineering Achievement Award
- 2017 MIT Technology Review: Innovator Under 35 Asia
- 2017 Forbes 30 Under 30 Asia: Healthcare
- 2017 National Research Foundation Fellowship
- 2016 NUS Young Investigator Award
- 2012 National Defense Engineering & Science Graduate Fellowship (USA)

SELECTED PUBLICATIONS

Published >70 papers in top venues in electrical engineering, physics, biomedical engineering, and materials science that have been cited >12k times. Link to [Google Scholar](https://scholar.google.com/citations?user=J5).



- J5 R. Lin*, H. Kim*, S. Achavananthadith, S. A. Kurt, S. C. C. Tan, H. Yao, B. C. K. Tee, J. L. W. Lee, J. S. Ho, "Wireless battery-free body sensor networks using near-field-enabled clothing," *Nature Communications*, 11, 444 (2020). [Listed as a Top 50 Physics Article of 2020. [Press coverage](#).]



- J4 Z. Dong, Z. Li, F. Yang, C. W. Qiu, J. S. Ho, "Sensitive readout of implantable microsensors using a wireless system locked to an exceptional point," *Nature Electronics*, 2, 335-342 (2019). [Featured in News & Views. [Press coverage.](#)]
- J3 X. Tian*, P. M. Lee*, Y. J. Tan, T. L. Y. Wu, H. Yao, M. Zhang, Z. Li, K. A. Ng, B. C. K. Tee, J. S. Ho, "Wireless body sensor networks based on metamaterial textiles", *Nature Electronics*, 2, 242-251 (2019). [Featured on the cover and in News & Views. [Press coverage.](#)]
- J2 A. Bansal, F. Yang, X. Tian, Z. Yong, J. S. Ho, "In vivo wireless photonic photodynamic therapy," *PNAS*, 115 (7), 1469-1474 (2018).
- J1 D. R. Agrawal, Y. Tanabe, D. Weng, S. Liao, Z. Zhen, Z. Zhu, C. Sun, Z. Dong, F. Yang, H. F. Tse, A. S. Y. Poon, and J. S. Ho, "Conformal phased surfaces for wireless powering of bioelectronic microdevices," *Nature Biomedical Engineering*, 1, 0043 (2017). [Featured in Editorial and News & Views.]

SELECTED PATENTS

Inventor on 3 granted patents and 4 pending patents, 4 of which are licensed.

- P5 *International Patent Pending PCT/SG2019/050515*. "Radio-Wave Confinement On Metamaterial Textiles For Wireless Sensor Networking". Filed Nov 21, 2018.
- P4 *International Patent Pending PCT/SG2019/050037*. J. S. Ho, Y. Zhang, A. Bansal, F. Yang, "Photodynamic Therapy Devices, Systems and Methods". Filed Feb 2, 2018. [Licensed to Incando Therapeutics]
- P3 *US Patent No. 10,594,166*. "Efficient wireless power transfer with flat, flexible lens". Filed Sep 28, 2015; issued Mar 17, 2020. [Licensed to Boston Scientific (2016–2017) and Neuspera Medical (2017–Present), >\$35k licensing revenue]
- P2 *US Patent No. 10,434,329*. "Autofocus wireless power transfer to implantable devices in freely moving animals". Filed Mar 25, 2015; issued Oct 8, 2019. [Licensed to MapLight Therapeutics (2019–Present)]
- P1 *US Patent No. 9,687,664*. "Wireless midfield systems and methods". Filed Sep 16, 2014; issued Jun 27, 2017. [Licensed to Neuspera Medical (2014–Present), 6 continuations granted and 5 pending, >\$130k licensing revenue]

PROFESSIONAL SERVICE

- 2025-Present Editorial Board, *Physical Review Applied* and *Materials Today Electronics*
- 2015-Present Reviewer for *Nature* series journals, *Science* series journals, *Physical Review* series journals, *Advanced Materials* series journals, *PNAS*, *IEEE Transactions on Antennas and Propagation*, *IEEE Microwave Theory and Techniques*, *IEEE Transactions on Biomedical Circuits and Systems*, *The Lancet Digital Health*, *ACS Nano*, and others.
- 2015-Present Served in an organizing role in the following conferences: ICBME2015, AWPT2017, IEEE APCAP2017, ICBME2019, IEEE APMC2019, IEEE WPTC 2020, IEEE AP-S/URSI 2021, IEEE BHI/BSN 2021, IEEE WPW 2022, IEEE IMBioC 2022, MRS Spring 2024.

SELECTED GRANTS

I was the principal investigator on grants totaling over USD\$3.5 million.

- 2017-2022 "Small-scale Wireless Devices for Bioelectronic Therapies"
National Research Foundation Fellowship (USD\$1.8M)
- 2017-2022 "Remote-controlled Photo-Therapy with Small-scale Wireless Bioelectronics", Ministry of Education, Tier 3 Grant (USD\$900k)